

SMART IoT-BASED IRRIGATION SYSTEM

Project Portfolio

Developed By

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Figure 1: Completed Smart IoT-Based Irrigation System Prototype

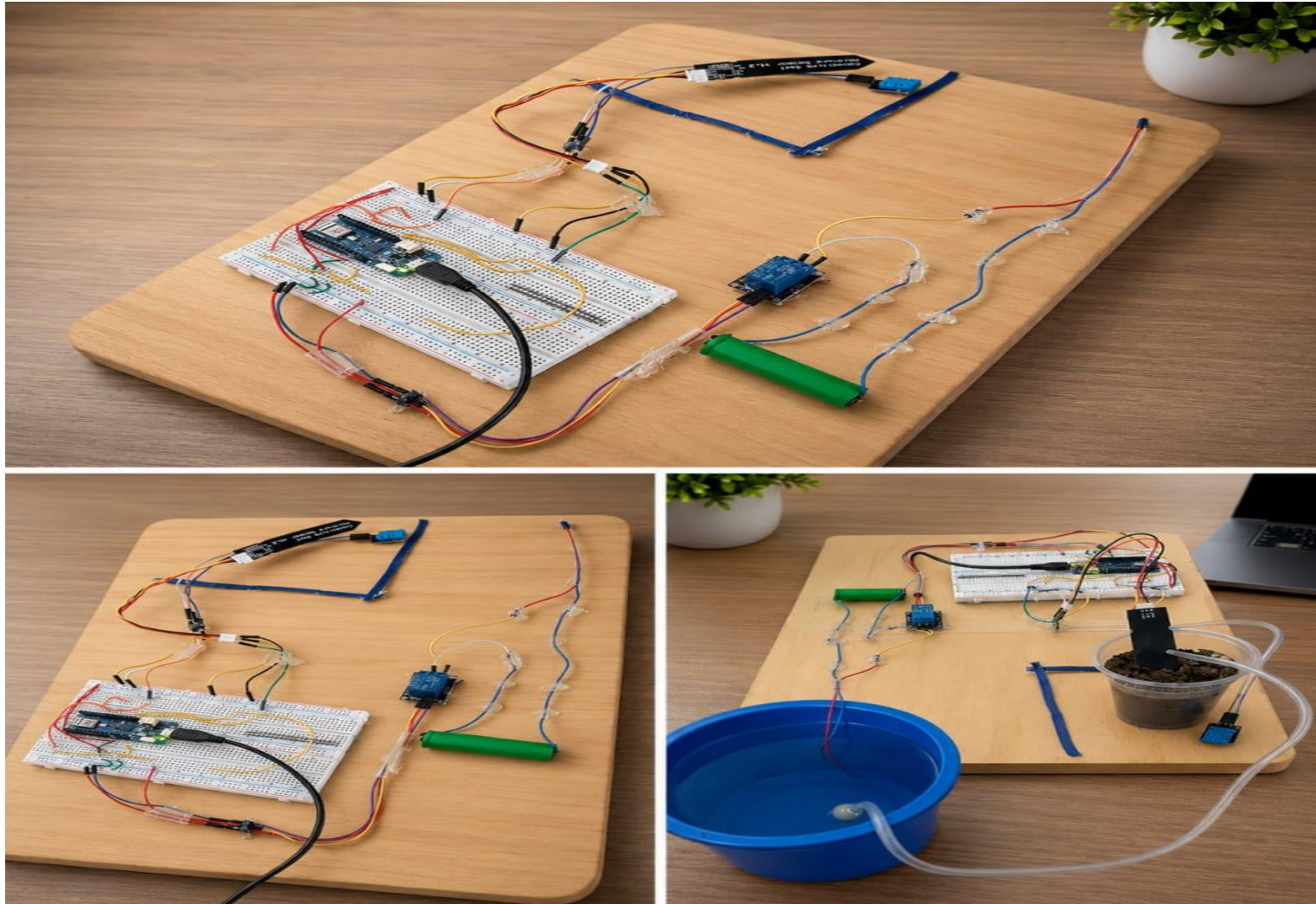


Table of Contents

1. *Project Summary*
2. *Project Overview*
3. *Problem Statement*
4. *Project Objectives*
5. *Components Used*
6. *System Architecture*
7. *Methodology*
8. *Working Principle*
9. *Results and Testing*
10. *Skills Demonstrated*
11. *Future Improvements*
12. *Conclusion*
13. *List of Figures*
14. *Appendix*

Project Summary

Items	Details
➤ <i>Project Title</i>	➤ <i>Smart IoT-Based Irrigation System</i>
➤ <i>Project Area</i>	➤ <i>Smart Agriculture & IoT</i>
➤ <i>Microcontroller</i>	➤ <i>Arduino MKR WiFi 1010</i>
➤ <i>Sensors</i>	➤ <i>Capacitive Soil Moisture Sensor, DHT11</i>
➤ <i>Connectivity</i>	➤ <i>WiFi & Arduino IoT Cloud</i>
➤ <i>Actuator</i>	➤ <i>Water Pump Via Relay</i>
➤ <i>Status</i>	➤ <i>Successfully Implemented</i>
➤ <i>Developer</i>	➤ <i>Muhammed Manjang</i>

1. Project Overview

The Smart IoT-Based Irrigation System is an automated agricultural solution designed to monitor soil moisture levels and control irrigation based on real-time environmental conditions.

The system combines *embedded systems, sensors, wireless communication, and automation technologies* to optimize water usage and reduce manual intervention in irrigation practices.

Through Internet of Things (IoT) technology, users can remotely monitor environmental conditions and irrigation status using a mobile application.

2. Problem Statement

Agriculture relies heavily on proper irrigation management. Traditional irrigation methods often involve manual watering, which can lead to:

- Excessive water consumption
- Underwatering or overwatering of crops
- Increased labor requirements
- Reduced agricultural productivity

An intelligent irrigation system is therefore needed to automate watering decisions based on actual soil conditions.

3. Project Objectives

- ❖ Main Objective
 - To design and implement a Smart IoT-Based Irrigation System capable of automatically controlling water supply based on soil moisture conditions & remote control.
- ❖ Specific Objectives
 - Monitor soil moisture levels in real time.
 - Measure environmental temperature and humidity.
 - Automate irrigation processes.
 - Enable remote monitoring through IoT technology.
 - Improve water efficiency.
 - Demonstrate practical implementation of IoT in agriculture.

4. Components Used

Components	Purpose
➤ Arduino MKR WiFi 1010	➤ Main controller

➤ Capacitive Soil Moisture Sensor	➤ Soil Moisture Measurement
➤ DHT11	➤ Temperature & Humidity Monitoring
➤ 5V Relay	➤ Switching Device
➤ Water Pump	➤ Irrigation
➤ BreadBoard	➤ Circuit Assembly
➤ 3.7V Battery	➤ Power Supply
➤ Jumper Wires	➤ Connection

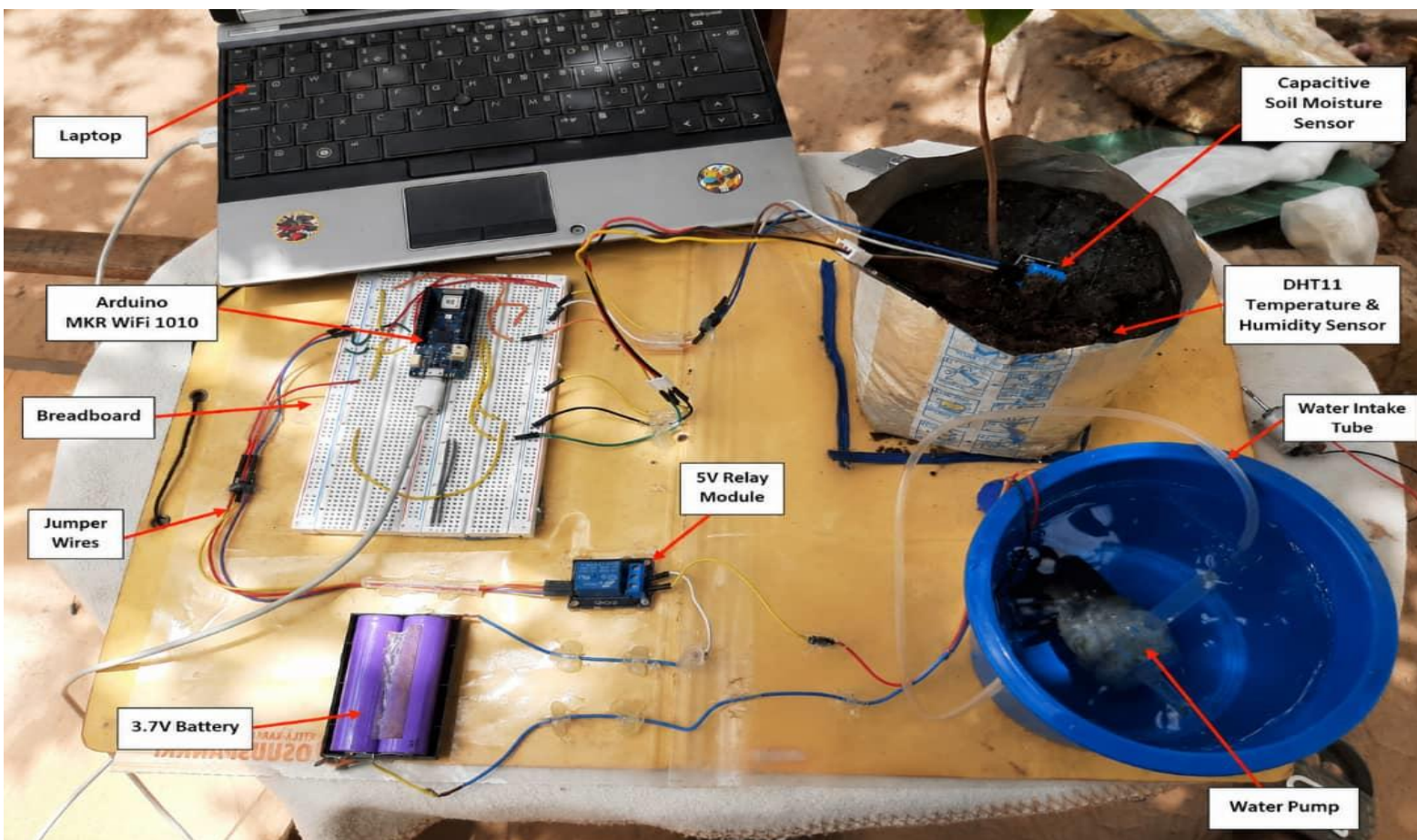


Figure 2: Hardware Components Used

5. System Architecture

The system consists of three major sections:

- ❖ Sensing Section
 - Collects environmental and soil data using:
 - Capacitive Soil Moisture Sensor
 - DHT11 Sensor
- ❖ Processing Section

- The Arduino MKR WiFi 1010 processes sensor data and determines whether irrigation is required.
- ❖ Actuation Section
- The relay module controls the water pump according to the decisions made by the controller.

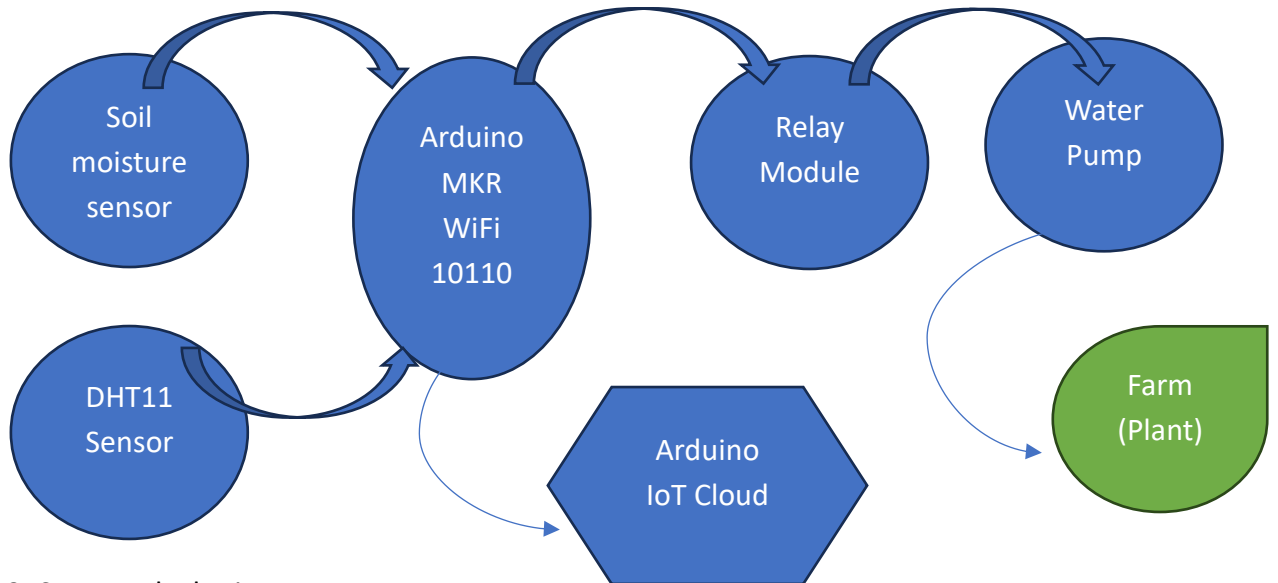


Figure 3: System Block Diagram

6. Methodology

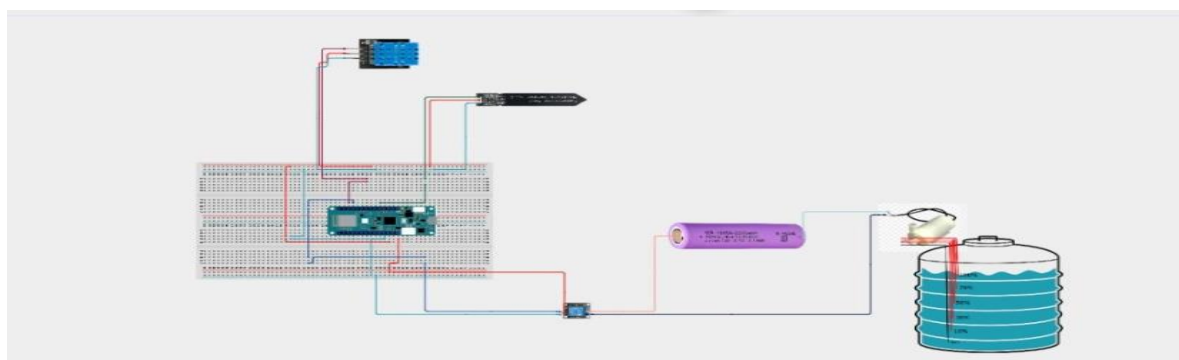
Research and Planning

Existing smart irrigation systems and IoT technologies were studied to identify suitable design approaches.

Hardware Assembly

All components were connected according to the designed circuit.

Figure 4: Hardware Assembly



Software Development

❖ *The Arduino MKR WiFi 1010 was programmed using Arduino IDE to:*

- Read sensor data
- Compare moisture values against threshold levels
- Control the relay module
- Communicate with Arduino IoT Cloud

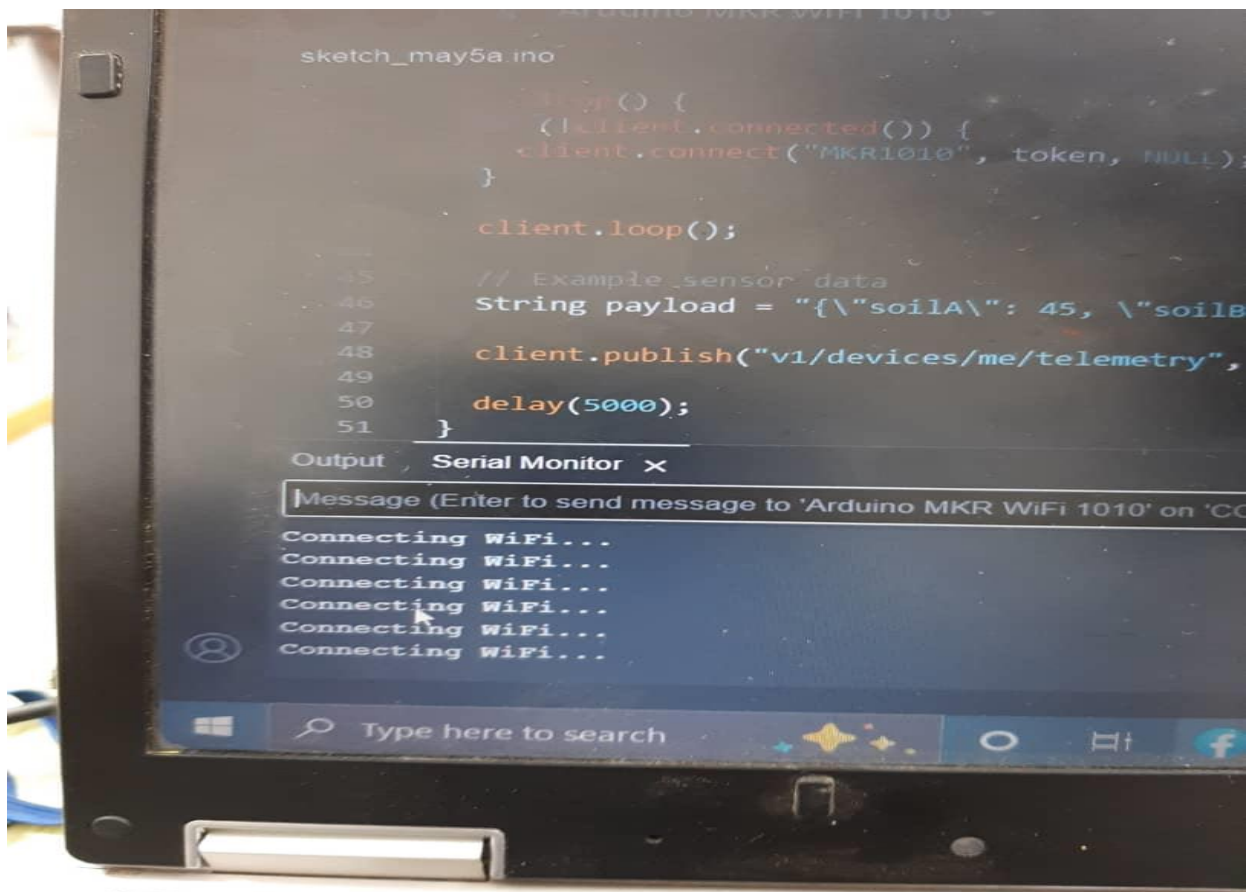
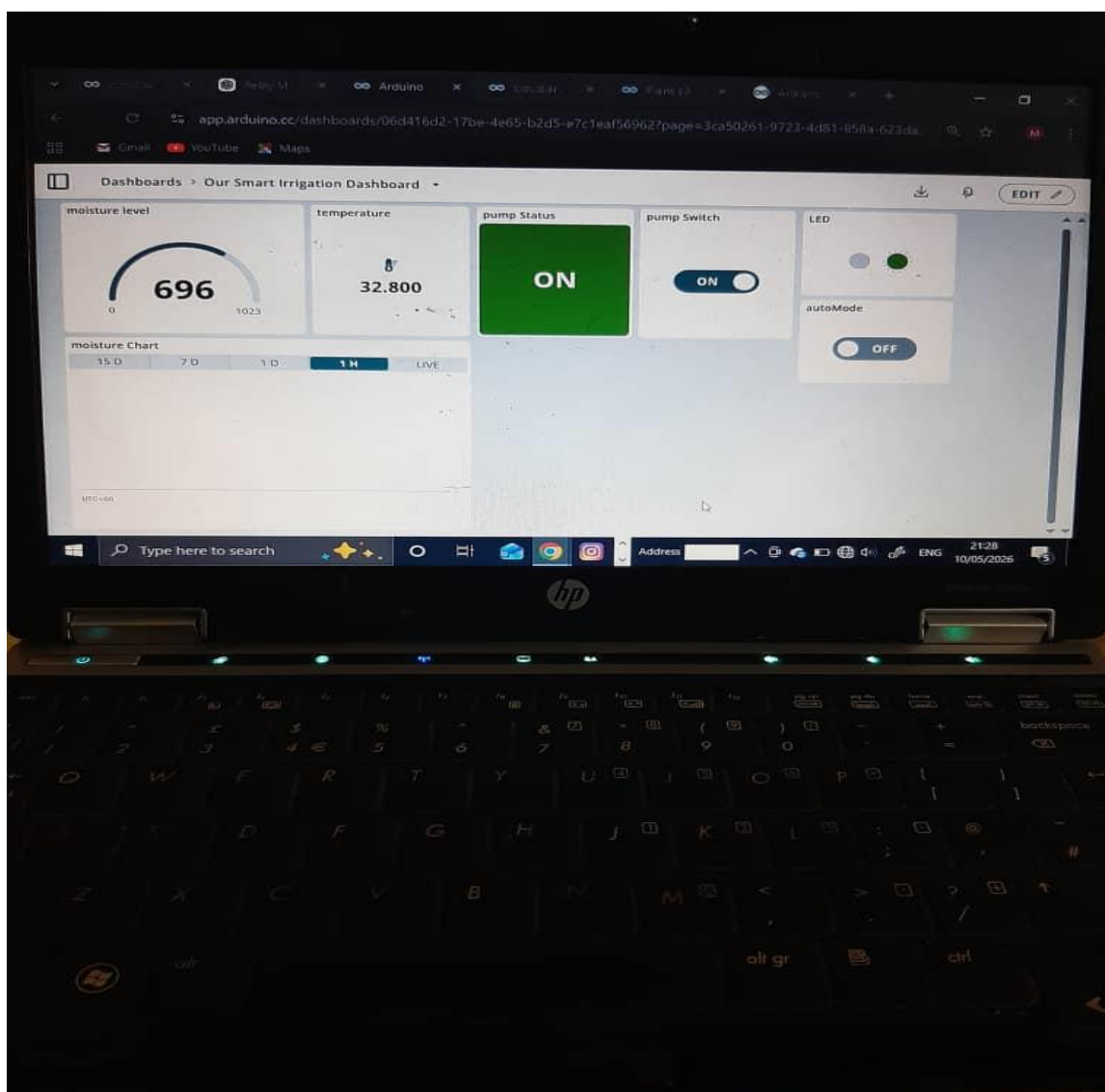


Figure 5: Arduino Program Development

IoT Integration

The Arduino IoT Cloud platform was configured to provide wireless communication and remote monitoring capabilities.

Figure 6: Arduino IoT Cloud Dashboard



7. Working Principle

The system operates according to the following sequence:

- Soil moisture sensor measures soil conditions.
- DHT11 measures temperature and humidity.
- Sensor readings are transmitted to the Arduino MKR WiFi 1010.
- The controller compares the moisture value with a predefined threshold.
- If the soil is dry:
 - Relay activates.
 - Water pump turns ON.
- If sufficient moisture is detected:
 - Relay deactivates.
 - Water pump turns OFF.
- Sensor readings are displayed on the IoT dashboard

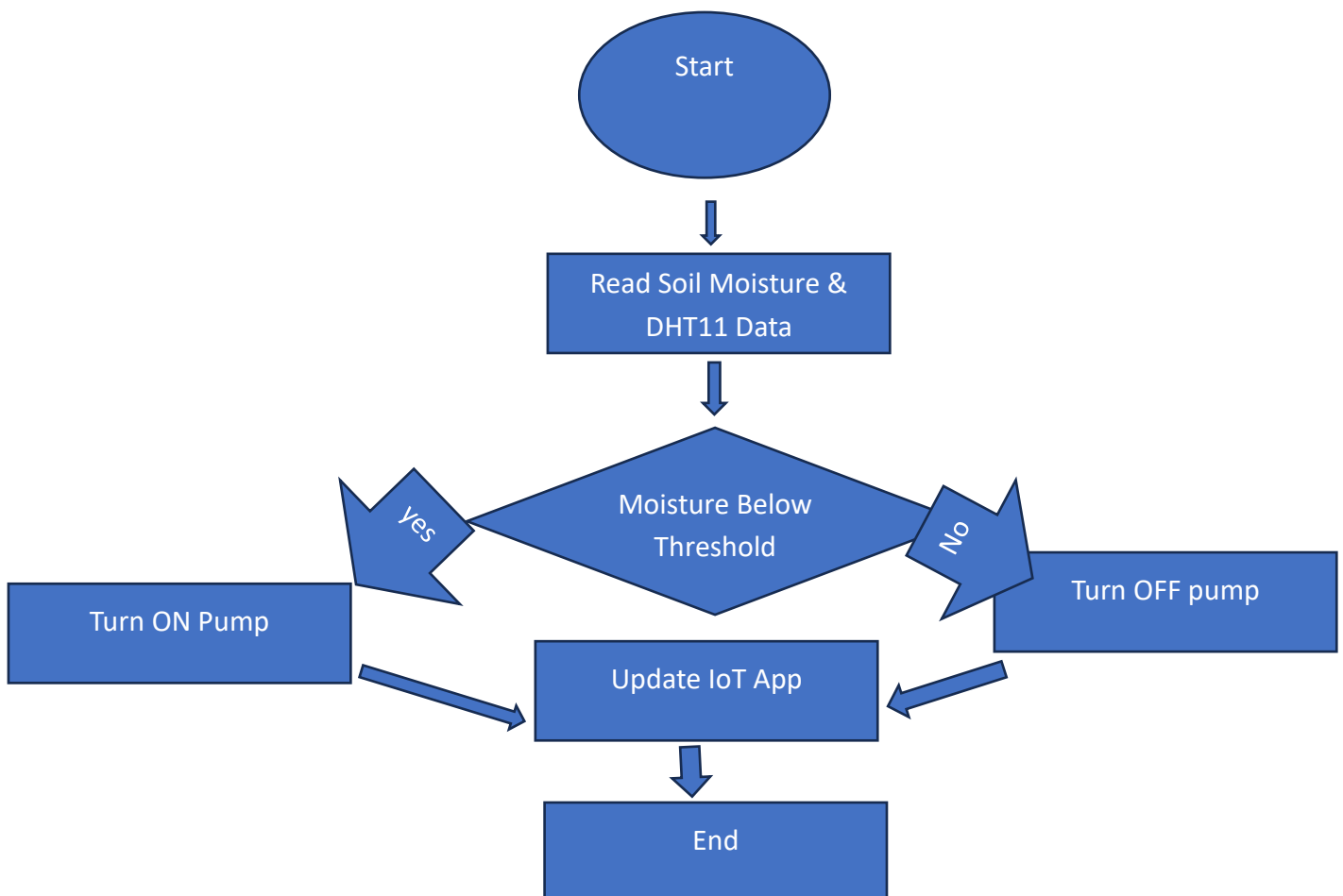


Figure 7: System Operation Flowchart

8. Results and Testing

The system was tested under different soil conditions.

Test Condition	Pump Status	Result
Dry Soil	ON	Successful
Moist Soil	OFF	Successful
Remote Monitoring	Operational	Successful

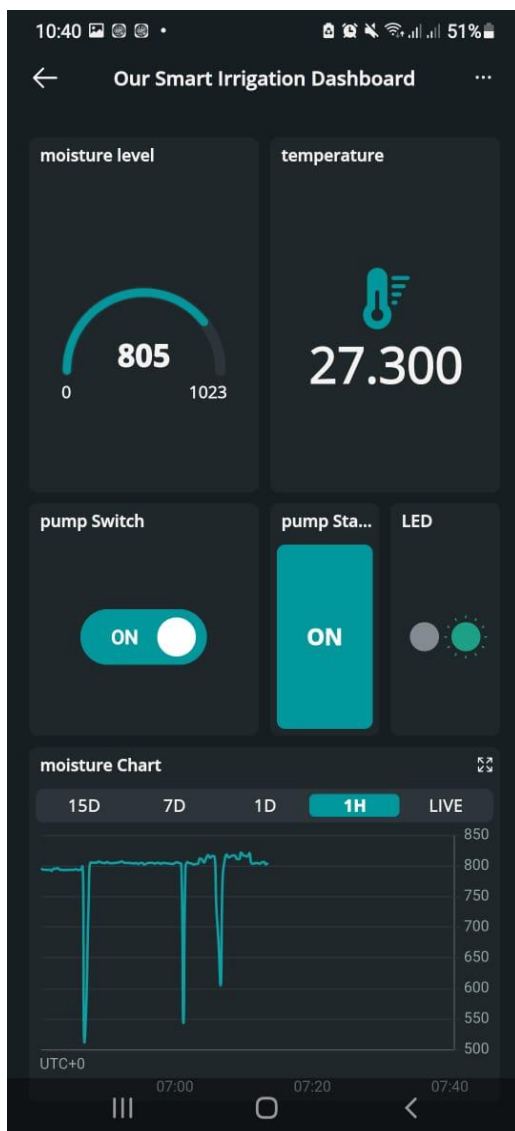
❖ Key Achievements

- ✓ Automatic irrigation achieved
- ✓ Real-time monitoring achieved
- ✓ Water conservation improved
- ✓ Remote accessibility achieved



Figure 8: Irrigation in Operation

Figure 9: Mobile Monitoring Interface



9. Skills Demonstrated

- ❖ Technical Skills
 - Arduino Programming
 - Embedded Systems Design
 - IoT Development
 - Sensor Integration
 - Wireless Communication
 - Circuit Design
 - Automation Systems
- ❖ Professional Skills
 - Problem Solving
 - Project Planning
 - Technical Documentation
 - System Testing
 - Research and Development

10. Future Improvements

- ❖ Potential enhancements include:
 - Solar-powered operation
 - Weather forecast integration
 - SMS notifications
 - Cloud data analytics
 - Multi-zone irrigation control
 - AI-based irrigation scheduling

11. Conclusion

The Smart IoT-Based Irrigation System successfully demonstrated the practical application of embedded systems and IoT technologies in agriculture.

The project achieved automatic irrigation control, remote monitoring, and efficient water management while providing valuable experience in embedded systems design, automation, wireless communication, and agricultural technology.

- ✓ This project highlights the potential of smart technologies to contribute to sustainable and efficient agricultural practices.

12. List of Figures

Figure 1: Completed Smart IoT-Based Irrigation System Prototype

Figure 2: Hardware Components Used

Figure 3: System Block Diagram

Figure 4: Hardware Assembly

Figure 5: Arduino Program Development

Figure 6: Arduino IoT Cloud Dashboard

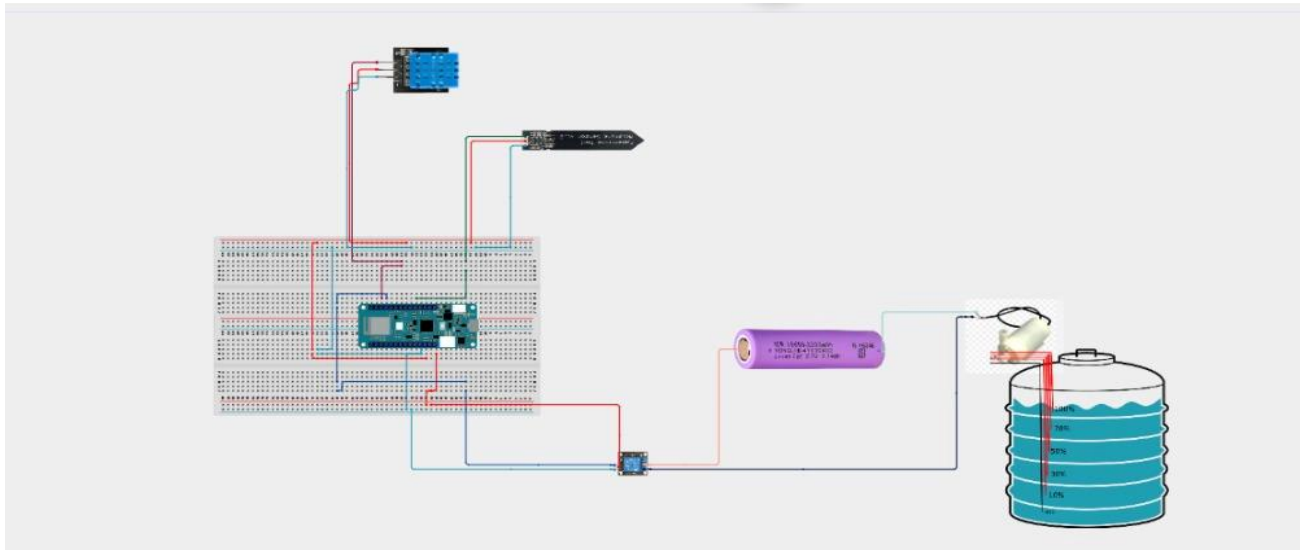
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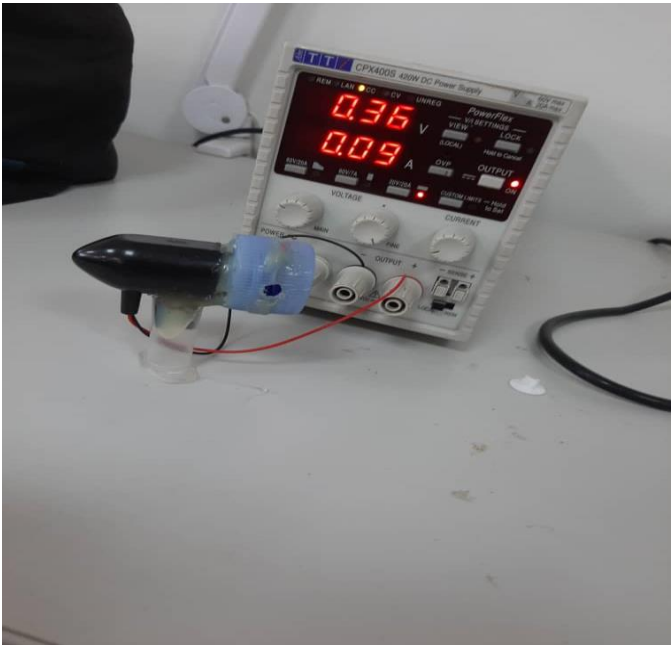
13. Appendix

Appendix A: Circuit Diagram



Appendix B: Source Code

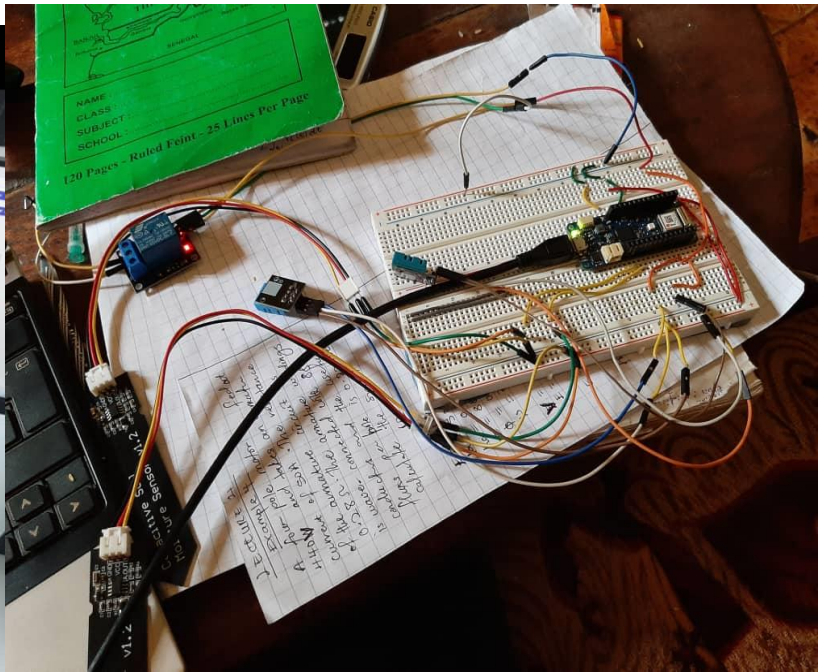
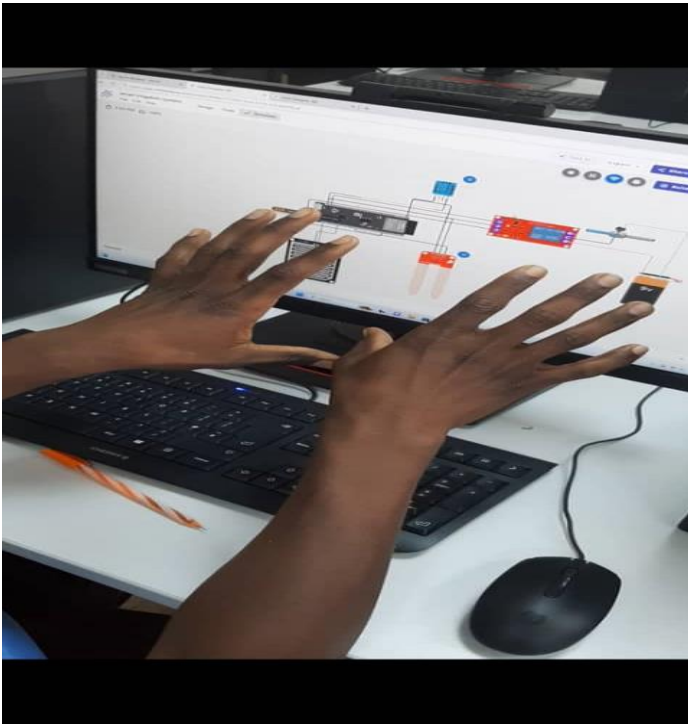
Appendix C: Additional Project Photographs



Design and construction of the water pump



Integrating the Arduino MKR WiFi 1010 with iot cloud



My initial circuit design phase



